

$$u(x_1, x_2, t)$$

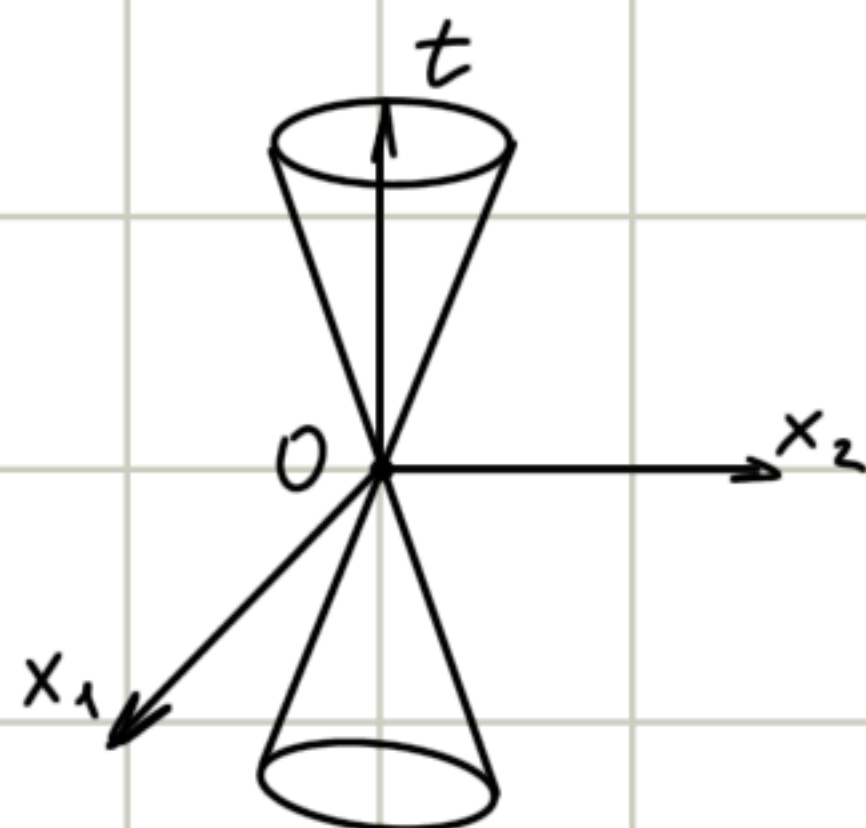
$$u_{x_1 x_1} + u_{x_2 x_2} - u_{tt} = 0$$

$$S(x_1, x_2, t) = 0$$

$$(S'_{x_1})^2 + (S'_{x_2})^2 - (S'_t)^2 = 0$$

$$x_1^2 + x_2^2 - t^2 = 0$$

$$S(x_1, x_2, t)$$



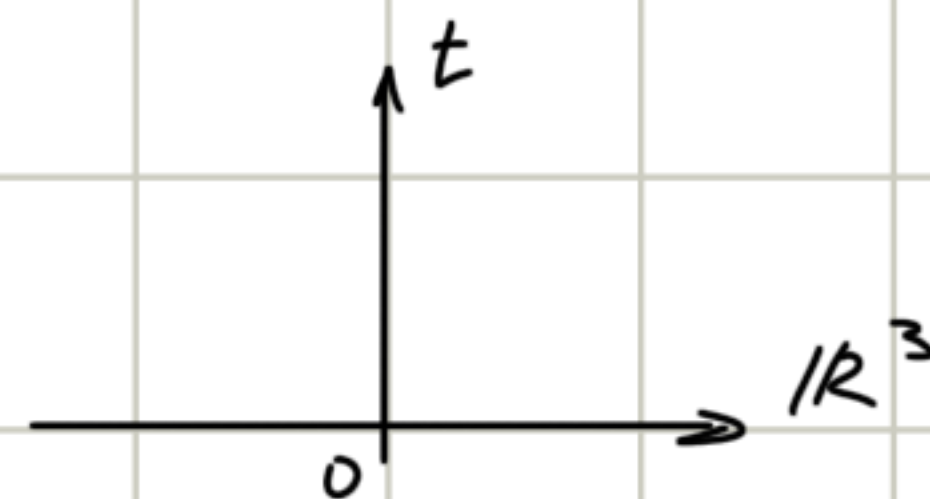
$$(x_1 - x_1^0)^2 + (x_2 - x_2^0)^2 = (t - t^0)^2$$

$$(0, 0, 0) \rightarrow (x_1^0, x_2^0, t^0)$$

Задача Коши для волн. ур. (анализ 3х простран. перемен.)

$$x = (x_1, x_2, x_3) \quad a \in \mathbb{R}, a > 0$$

$$u(x, t)$$



$$(1) u_{tt} - a^2 \Delta u = F(x, t), \quad x \in \mathbb{R}^3, \quad t > 0$$

$$(2) u|_{t=0} = \varphi(x), \quad x \in \mathbb{R}^3$$

$$(3) u_t|_{t=0} = \psi(x), \quad x \in \mathbb{R}^3$$

Опр.

Φ -а $u(x, t)$ абс. переносим (1), (2), (3):

$$1) u(x, t) \in C^2 (x \in \mathbb{R}^3, t \geq 0)$$

$$2) \text{ ур. (1), (2) и (3)}$$

предполагаем $u(x, t)$ абс. перен.

$$\Rightarrow \varphi(x) \in C^2(\mathbb{R}^3)$$

$$\psi(x) \in C^1(\mathbb{R}^3)$$

$$F(x, t) \in C(x \in \mathbb{R}^3, t > 0)$$

$$F(x, t) \equiv 0$$

$$a = 1$$

$$(1_0) u_{tt} - \Delta u = 0, \quad x \in \mathbb{R}^3, \quad t > 0$$

$$(2) u|_{t=0} = \varphi(x), \quad x \in \mathbb{R}^3$$

$$(3) u_t|_{t=0} = \psi(x), \quad x \in \mathbb{R}^3$$

th 3-ния решения

$$\exists \varphi(x) \in C^3(\mathbb{R}^3)$$

$$\psi(x) \in C^2(\mathbb{R}^2)$$

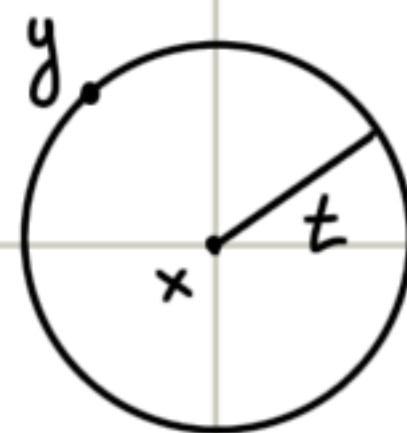
могда реш. з-ни (1), (2), (3) 3-ем и заг. ф-лой

$$u(x,t) = \frac{1}{4\pi t} \iint_{|y-x|=t} \psi(y) dS_y + \frac{\partial}{\partial t} \left(\frac{1}{4\pi t} \iint_{|y-x|=t} \varphi(y) dS_y \right)$$

формула Кирхгофа

поверх. шм-а 1^{го} рода
по сфере с центром
в м. x и R=t

$$y = (y_1, y_2, t)$$

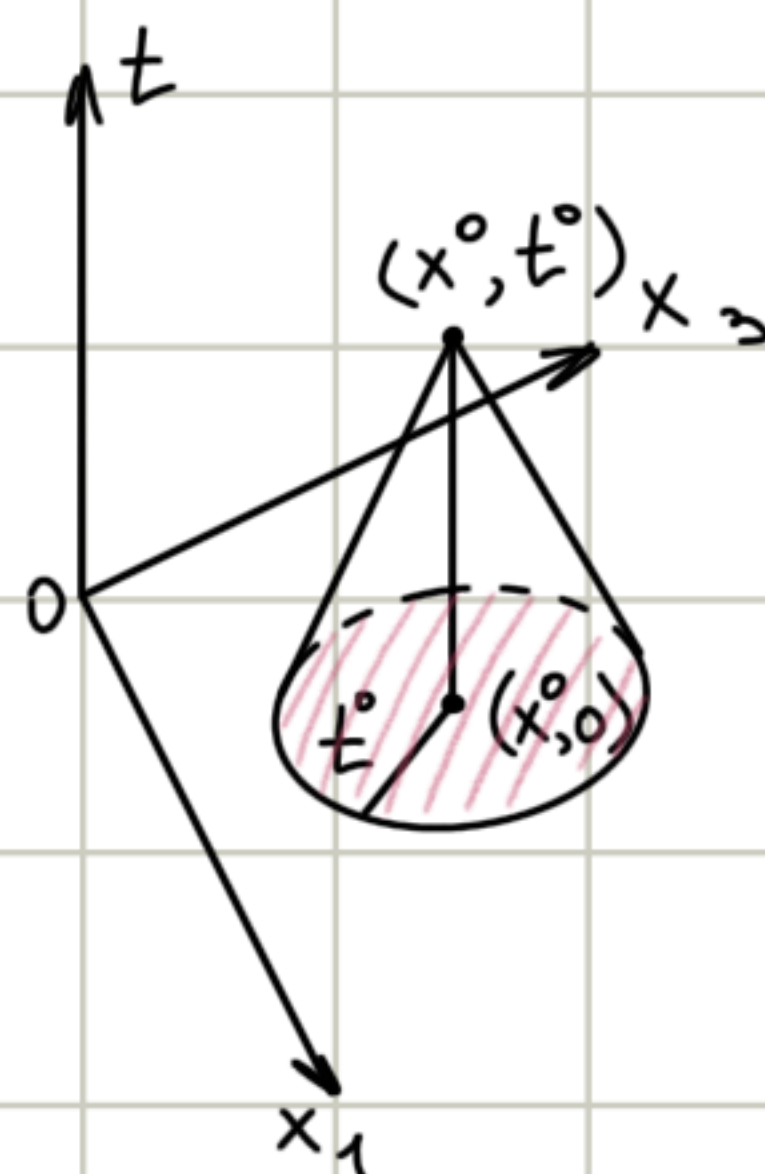
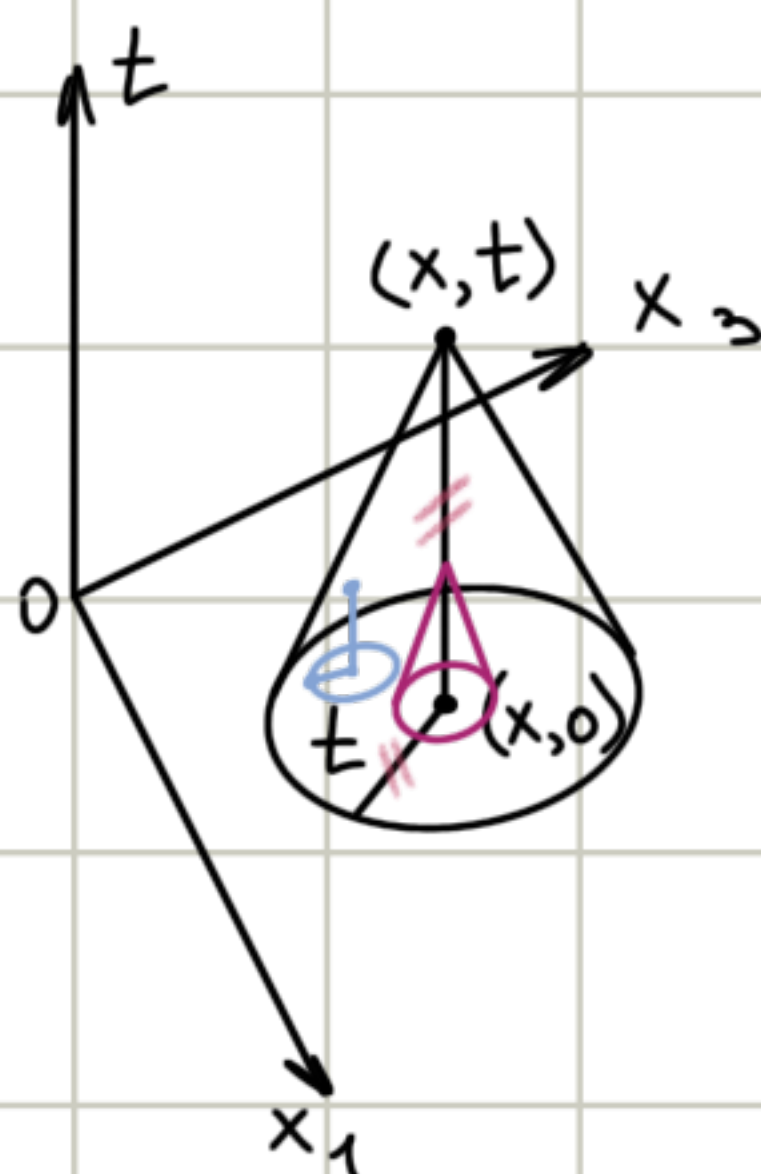


$$g(x)$$

$$t > 0, x \in \mathbb{R}^3$$

$$\frac{1}{4\pi t} \iint_{|y-x|=t} g(y) dS_y = A_g(x,t)$$

$$x^0 \in \mathbb{R}^3, t^0 > 0 \mapsto (x^0, t^0) \in \mathbb{R}^4$$



$$\overline{Q}_{x^0, t^0} = \{(x, t): \begin{matrix} t=0 \\ (x_1-x_1^0)^2 + (x_2-x_2^0)^2 + (x_3-x_3^0)^2 \leq (t^0)^2 \end{matrix}\}$$

$$\overline{Q}_{x^0, t^0} = \{(x, t): \begin{matrix} (x_1-x_1^0)^2 + (x_2-x_2^0)^2 + (x_3-x_3^0)^2 \leq (t-t^0)^2 \\ 0 \leq t \leq t^0 \end{matrix}\}$$

$$g(x), x \in \overline{Q}_{x^0, t^0} \mapsto A_g(x, t), (x, t) \in \overline{Q}_{x^0, t^0}$$

$$A_g(x, t) = \frac{1}{4\pi t} \iint_{|y-x|=t} g(y) dS_y$$

$$y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \mapsto \bar{\eta} = \begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix}$$

$$\begin{cases} y_1 - x_1 = \eta_1 t \\ y_2 - x_2 = \eta_2 t \\ y_3 - x_3 = \eta_3 t \end{cases}$$

$$|\bar{\eta}| = 1$$

$$\begin{aligned} |y-x| &= t \\ |y-x| &= |\eta t| = t |\eta| \\ |\eta| &= 1 \end{aligned}$$

$$g(y) dS_y = \frac{t^2}{4\pi t} \iint_{|\eta|=1} g(x_1 + \eta_1 t, x_2 + \eta_2 t, x_3 + \eta_3 t) dS_\eta$$

x_1, x_2, x_3, t - параметры $\overline{Q}_{x^0, t^0}$

$\eta_1, \eta_2, \eta_3 \in \{|\eta|=1\}$ перем. шм.

$$g(y_1, y_2, y_3) = g(x_1 + \eta_1 t, x_2 + \eta_2 t, x_3 + \eta_3 t)$$

производные $A_g(x, t) \rightarrow$ по t

по x_1, x_2, x_3

$$\frac{\partial}{\partial x_1} (A_g(x, t)) = \frac{t}{4\pi} \iint_{|y-x|=t} \frac{\partial g}{\partial y_1}(\dots) \cdot 1 dS_y = \frac{1}{4\pi t} \iint_{|y-x|=t} \frac{\partial g(y)}{\partial y_1} dS_y = A_{\frac{\partial g}{\partial y_1}}(x, t)$$

$$\frac{\partial^2}{\partial x_1^2} (A_g(x, t)) = \frac{t}{4\pi} \iint_{|y-x|=t} \frac{\partial^2 g}{\partial y_1^2}(\dots) dS_y = A_{\frac{\partial^2 g}{\partial y_1^2}}(x, t)$$

$$\frac{1}{4\pi t} \iint_{|y-x|=t} (\bar{\nabla} g, \bar{\eta}) dS_y = \frac{1}{4\pi t} \iiint_{|y-x|<t} \Delta g(y) dy = \frac{1}{4\pi t} \int_0^t dr \left(\iint_{|y-x|=r} \Delta g(y) dS_y \right)$$

$$\Delta (A_g(x, t)) = A_{\Delta g}(x, t)$$

$$\frac{\partial}{\partial t} (A_g(x, t)) = \frac{1}{4\pi} \iint_{|y-x|=1} g(\dots) dS_y + \frac{t}{4\pi} \iint_{|y-x|=1} \left(\frac{\partial g(\dots)}{\partial y_1} \eta_1 + \frac{\partial g(\dots)}{\partial y_2} \eta_2 + \frac{\partial g(\dots)}{\partial y_3} \eta_3 \right) dS_y$$

$(\bar{\nabla} g(\dots), \bar{\eta})$

$$\bar{\nabla} g = \begin{pmatrix} g_{y_1} \\ g_{y_2} \\ g_{y_3} \end{pmatrix}; \quad \bar{\eta} = \begin{pmatrix} \eta_1 \\ \eta_2 \\ \eta_3 \end{pmatrix}$$

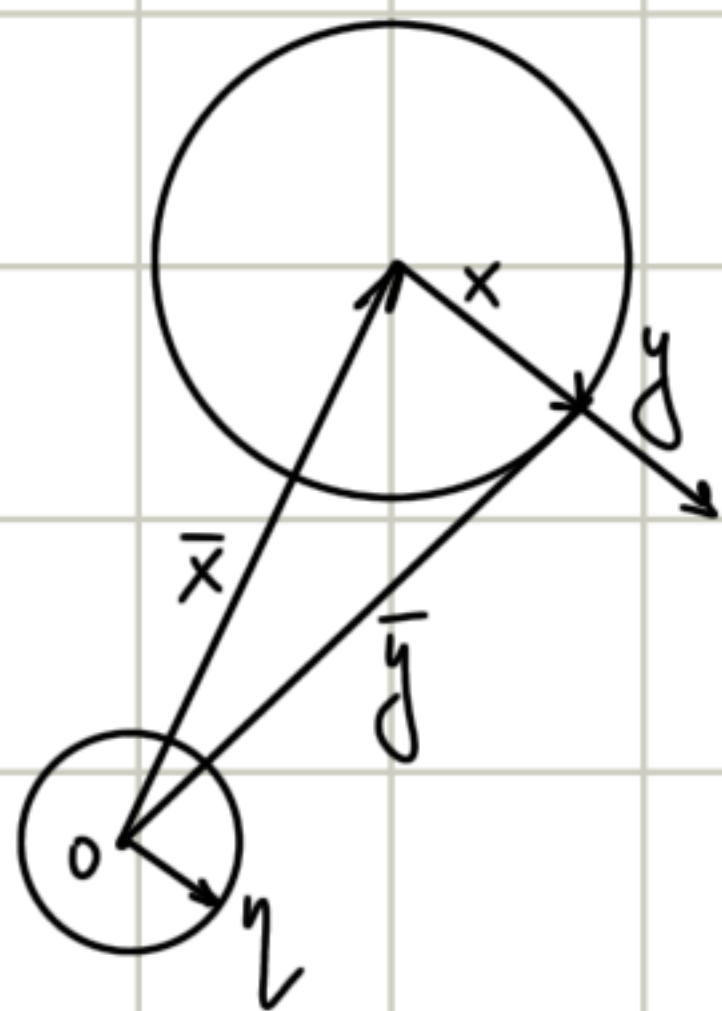

$$\frac{1}{4\pi} \iint_{|y-x|=1} g(x_1, x_2, x_3) dS_y = \frac{1}{4\pi} g(x_1, x_2, x_3) \iint_{|y-x|=1} 1 dS_y = g(x)$$



$$a(x) \in C^1(\bar{V})$$

$$\int \dots \int_V \operatorname{div} \bar{a}(x) = \iint_{\partial V} (\bar{a}, \bar{n}) dS$$

ф-ла Остроградского



$$\bar{y} - \bar{x} = t \cdot \bar{\eta}$$

$$g \in C^2, \quad \nabla \bar{g} \in C^1$$

$$\bar{\nabla} g = \begin{pmatrix} g_{y_1} \\ g_{y_2} \\ g_{y_3} \end{pmatrix}$$

$$\operatorname{div}(\bar{\nabla} g) = g_{y_1 y_1} + g_{y_2 y_2} + g_{y_3 y_3}$$

$$\frac{\partial}{\partial t} (A_g(x, t)) = \frac{1}{4\pi t^2} \iint_{|y-x|=t} g(y) dS_y + \frac{1}{4\pi t} \int_0^t dr \left(\iint_{|y-x|=r} \Delta g(y) dS_y \right)$$

"Сферический"

$$1 \quad A_g(x, t)|_{t=0} = 0$$

$$2 \quad \text{шаровые}$$

$$3 \quad \Delta (A_g(x, t)) = A_{\Delta g}(x, t)$$

$$4 \quad \frac{\partial}{\partial t} A_g(x, t)|_{t=0} = g(x)$$